



TKANMiXer: multi-scale hybrid KAN for long-term forecasting

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Abstract

Time series forecasting is extensively applied across diverse real-world domains, such as finance, power systems, and meteorology, yet its accuracy is often constrained by the inherent complexity of temporal data. While exogenous variables can enhance predictive performance, their inclusion introduces additional challenges by increasing the intricacy of the time series structure. Moreover, traditional Multi-Layer Perceptrons (MLPs) frequently fail to effectively capture hierarchical feature representations across multiple scales, further complicating time series modeling. To address these limitations, we propose TKANMiXer, a novel Kolmogorov-Arnold Network(KAN)-based framework designed for variable fusion and multi-scale decomposition. Our framework comprises two key components: (1) incorporating a seasonal-trend decomposition module on the basis of variable fusion, (2) the multi-layer KAN architecture hierarchically captures. Specifically, TKANMiXer addresses the complexity arising from exogenous variables by disentangling temporal patterns, multi-layer KAN architecture hierarchically captures multi-scale temporal dependencies, significantly improving forecasting accuracy. Extensive experiments demonstrate that TKANMiXer achieves state-of-the-art performance in long-term forecasting tasks across multiple benchmark datasets.

Keywords Time series prediction · Deep learning · Attention mechanism · Seasonal trend decomposition · Kolmogorov-Arnold Network

1 Introduction

Time series forecasting serves as a fundamental component in real-world applications. By modeling time series data, temporal patterns can be extracted from historical records

to enable accurate future trend prediction, thereby providing essential data-driven support for decision-making processes. This capability is particularly valuable in high-stakes domains including finance Zhou et al. (2025), transportation Ali et al. (2024), power systems Weron (2014), and meteorology Li et al. (2025); Ali et al. (2021). Nevertheless, the inherent complexity of real-world data, marked by non-stationarity and dynamic external influences like economic policies and demographic changes, presents substantial challenges to the robustness and generalization of forecasting models.

External factors significantly influence time series forecasting. As depicted in Figure 1, contemporary forecasting methodologies primarily comprise two paradigms: univariate endogenous forecasting that depends exclusively on historical observations, and exogenous-assisted forecasting that systematically integrates multi-dimensional external variables. The exogenous-assisted paradigm markedly improves model completeness and predictive precision through the incorporation of environmental conditions, market indicators, and other contextual parameters. Modern deep learning frameworks Olivares et al. (2023) leverage exogenous variables as auxiliary inputs to improve the reliability of

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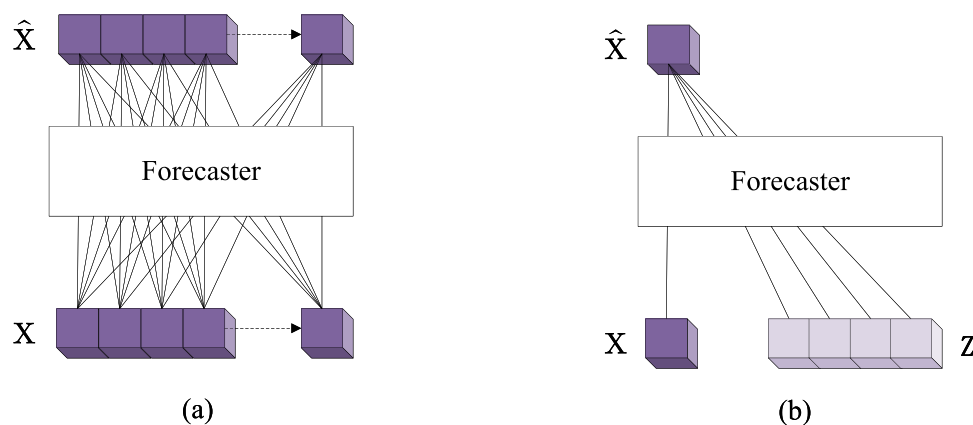
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Fig. 1 Comparative analysis of forecasting architectures: **(a)** Endogenous prediction via self-attention mechanisms ($X \rightarrow \hat{X}$), **(b)** Exogenous-fused prediction through cross-attention schemes ($X + Z \rightarrow \hat{X}$), where X denotes the input variable, Z represents auxiliary predictive information, and $\hat{X}$ is the predicted variable



endogenous predictions. Particularly, the TimeXer architecture Wang et al. (2024) demonstrates notable improvements in both accuracy and robustness via sophisticated fusion mechanisms between exogenous variables and temporal sequences, validating the efficacy of multi-source collaborative modeling. The integration of external determinants—including market equilibrium dynamics, geographical heterogeneity, and meteorological patterns—effectively mitigates the theoretical constraints of conventional time series approaches. Through multi-dimensional information fusion, these advanced methods more accurately capture the fundamental evolutionary mechanisms of temporal processes. However, while integrating exogenous variables strengthens the characterization of temporal dependencies, the increased system complexity poses significant challenges in identifying essential dynamic relationships.

The inherent complexity of time series data is substantially amplified by nonlinear interdependencies among their multivariate components in practical applications. This necessitates models capable of performing concurrent multiscale integration and discriminative feature extraction. Current methodologies address this challenge through domain-specific temporal decomposition techniques such as season-trend separation and frequency-domain analysis, which systematically decouple entangled temporal relationships while maintaining interpretability in complex time series modeling Zeng et al. (2023); Wang et al. (2023). Specifically, Autoformer Wu et al. (2021) introduces an adaptive decomposition architecture that explicitly models cross-scale feature interactions, demonstrating superior performance in capturing complex temporal dependencies. Building upon this, TimeMiXer Wang et al. (2024) proposes a hierarchical decomposition framework with cascaded fusion mechanisms, enabling effective modeling of dynamical patterns in complex time series. The method's capability to distill essential temporal variations through structured simplification provides critical feature representations for forecasting tasks, yielding significant improvements in both prediction accuracy and model generalizability.

KAN have recently emerged as a transformative approach for temporal modeling, utilizing their unique B-spline-based univariate function composition architecture. By decomposing complex target functions into combinations of univariate components, KANs achieve significantly enhanced prediction accuracy and interpretability compared to conventional methods, enabling more efficient time series representation. This innovative architecture demonstrates superior data-fitting capacity and adaptability on small datasets, outperforming standard multilayer perceptrons in temporal feature extraction. C-KAN Livieris (2024) dynamically optimizes training through adaptive edge activation mechanisms, improving gradient propagation efficiency to accelerate convergence. KARN Danish and Grolinger (2025) combines recurrent architectures with KAN's nonlinear transformation capability for effective modeling of complex temporal dependencies. KAN-AD Zhou et al. (2024) employs Fourier analysis for global pattern extraction while leveraging univariate decomposition for noise suppression in anomaly detection. The wavelet-enhanced variant Jama-dar et al. (xxxx) utilizes Morlet transforms for multi-scale analysis of histopathological features. These advancements collectively validate KAN's exceptional capability in modeling intricate temporal dynamics, particularly in capturing nonlinear patterns and long-range dependencies.

Despite demonstrating remarkable predictive performance and parameter efficiency Vaca et al. (2024), KAN encounter significant computational challenges, particularly their $\sim 10\times$ slower training speed compared to parameter-equivalent multilayer perceptrons Xu et al. (2024). This computational limitation has motivated substantial architectural innovations, such as wavelet-based B-spline substitutions Bozorgasl et al. (2024), Gaussian radial basis function implementations (DeepOKAN) Abueidda et al. (2025), and Chebyshev polynomial variants Sidharth et al. (2024). These methodological advancements collectively improve training efficiency while preserving the representational advantages of KANs, thereby facilitating their practical deployment. Motivated by these developments, this work

introduces enhanced KAN architectures designed to overcome the inherent limitations of MLPs in temporal modeling. By replacing conventional MLP components, our method aims to significantly improve long-term forecasting accuracy in time series analysis.

We propose TKANMiXer, a novel network architecture designed to enhance time-series forecasting. The model integrates endogenous and exogenous variables to improve information representation, employs multi-scale sampling to decompose time series into fine and coarse-grained subsequences, and applies season-trend decomposition to disentangle complex patterns, reducing interference from external noise. Based on a hierarchical multi-scale fusion framework, TKANMiXer progressively extracts features at different scales and utilizes hybrid KAN layers to capture cross-cycle dependencies effectively. By exploiting the complementary strengths of different temporal granularities, the model achieves significant improvements in long-term forecasting accuracy. The main contributions of this paper are summarized as follows:

- We propose a novel approach that integrates variable fusion with seasonal-trend decomposition to address complex temporal variations caused by combining endogenous and exogenous variables. This method effectively reduces external interference while optimizing time-series feature representation.
- We design a multi-layer hybrid KAN architecture that performs targeted feature extraction for sub-sequences at different granularities, significantly enhancing the model's capability to capture complex multi-scale temporal dependencies.
- Comprehensive experiments on widely-used benchmark datasets (ETT, Exchange, DJI, FCHI, and SP500) demonstrate that TKANMiXer outperforms existing conventional methods in long-term forecasting tasks, establishing new state-of-the-art performance.

2 Related work

2.1 Time series forecasting models

In recent years, deep learning methods have demonstrated significant advances in time series forecasting, with the core challenge being accurate modeling of complex temporal dependencies in sequential data. Current approaches include RNN-based methods Salinas et al. (2020); Sherstinsky (2020), CNN-based architectures Wang et al. (2023, 2024), MLP-based techniques Challu et al. (2023), and Transformer-based frameworks Vaswani et al. (2017); Nie et al. (2022); Zhou et al. (2022). LSTM networks Yu et al.

(2019) process sequential data through recursive connections to capture temporal relationships, while CNN-based approaches like TimesNet Wu et al. (2022) employ temporal convolutional kernels to identify patterns in time series data. The DLinear model Zeng et al. (2023) further demonstrates that a simple MLP-based design can achieve competitive forecasting performance with superior computational efficiency. In multivariate forecasting, iTransformer Liu et al. (2023) effectively captures inter-variable dependencies by applying attention mechanisms to tokenized time series. Since its introduction, the Transformer architecture has become particularly influential for long-term forecasting tasks. Collectively, these diverse approaches have advanced the field through complementary innovations in temporal modeling, long-range dependency capture, and computational efficiency.

2.2 Application of variable fusion in time series

The integration of exogenous and endogenous variables in time series analysis typically involves incorporating external factors—such as weather conditions and economic indicators—as supplementary input features. This process is commonly implemented through multimodal fusion or attention mechanisms to enhance the model's capacity for capturing complex dynamical behaviors, thereby improving prediction accuracy. Traditional methods, such as SARIMAX Vagropoulos et al. (2016), jointly model the relationship between exogenous and endogenous variables while preserving the autoregressive properties of the endogenous series. More recently, TimeXer Wang et al. (2024) has demonstrated the effective incorporation of external information into endogenous representations using cross-attention mechanisms, particularly in scenarios where future exogenous values are available during inference. Nevertheless, in practical applications, the inclusion of external information frequently increases the complexity of temporal patterns, presenting substantial challenges for forecasting models.

2.3 Multi-scale analysis of time series

Time series multi-scale analysis utilizes techniques including moving averages and STL decomposition Cleveland et al. (1990) to separate multi-scale dynamic patterns from complex temporal structures. The Autoformer framework pioneered the integration of moving average-based time series analysis into deep learning models, decomposing complex temporal patterns into seasonal and trend components to improve forecasting performance Zeng et al. (2023); Zhang et al. (2022). MICN extended this methodology by decomposing input sequences into seasonal and trend subsequences while incorporating multi-scale local and global

context modeling to better capture long-term dependencies in complex time series Wang et al. (2023). TimesNet Wu et al. (2022) employs multi-period analysis to separate mixed temporal variations into distinct components with varying cycle lengths. More recently, TimeMiXer Wang et al. (2024) has demonstrated enhanced modeling of complex temporal relationships through its multi-scale decomposition architecture, which implements a hybrid mechanism to differentiate between the extraction of historical information and the prediction of future values.

2.4 Application of kan in time series

The application of KAN in time series analysis is grounded in the Kolmogorov-Arnold representation theorem, which demonstrates that multivariate continuous functions can be approximated through compositions of univariate functions Liu et al. (2024). The fundamental innovation of KANs lies in replacing the fixed activation functions of traditional multilayer perceptrons (MLPs) with learnable activation functions at network nodes. This design provides enhanced function approximation capacity and structural adaptability, enabling superior generalization performance on small-scale datasets while maintaining high interpretability. These advantages position KANs as a promising alternative to MLPs for specific applications Genet et al. (2024); Genet and Inzirillo (2024); Lobanov et al. (2024); Wu et al. (2024). However, the original KAN implementation faced computational efficiency limitations due to the inherent complexity of spline functions. This challenge motivated the development of FastKAN, which incorporates optimized third-order spline functions to substantially improve training speed Li (2024). FastKAN thus preserves the approximation capabilities of KANs while overcoming previous computational bottlenecks, thereby extending the model's applicability to broader practical scenarios.

3 TKNMiXer

We propose the TKNMiXer architecture, which initially integrates endogenous and exogenous variables via a cross-attention mechanism. The input sequence is then downsampled into multiple subsequences, each of which undergoes seasonal-trend decomposition and mixing. Finally, a multi-layer hybrid KAN architecture is employed to resolve feature-related challenges in time-series forecasting.

3.1 Exogenous information fusion

In time series forecasting tasks, intrinsic correlations often exist among different feature variables. To effectively model

such relationships, we treat the endogenous time series to be predicted as the primary modeling target, while considering other relevant feature variables as exogenous factors that influence the dynamics of the endogenous series. These exogenous variables provide rich contemporaneous contextual information for historical observations of the target variable, thereby enhancing the model's understanding of past states. It is noteworthy that under this formulation, the model only requires historical exogenous variables during the inference phase, without relying on their future values. This approach not only aligns with practical deployment constraints but also avoids strong assumptions about future external information. TKNMiXer incorporates exogenous variable tokens into each endogenous sequence for information fusion, with its architecture detailed in Figure 2.

Given a historical sequence $\mathbf{X} \in \mathbb{R}^{T \times C}$, let $X_{1:T} = \{x_1, x_2, \dots, x_T\} \in \mathbb{R}^{T \times 1}$ represent the endogenous variable, where T denotes the number of time steps, C the number of variables, and x_i the value at the i -th time point. The exogenous variables are denoted as $Z_{1:T'} = \{Z_{1:T'}^{(1)}, Z_{1:T'}^{(2)}, \dots, Z_{1:T'}^{(C)}\} \in \mathbb{R}^{T' \times C}$, where T' denotes the number of time steps and $Z_{1:T'}^{(i)}$ represents the i -th exogenous variable. The goal of the forecasting model f is to predict the future S time steps $\mathbf{Y} = \{x_{T+1}, x_{T+2}, \dots, x_{T+S}\}$ based on both historical observations $X_{1:T}$ and the corresponding exogenous series $Z_{1:T'}^{(i)}$, which can be formulated as follows:

$$\mathbf{Y} = f(X_{1:T}, Z_{1:T'}^{(i)}). \quad (1)$$

In time series forecasting, endogenous variables constitute the target values for prediction, whereas exogenous variables represent external influencing factors. The endogenous variables are divided into multiple non-overlapping patches, each of which is transformed into temporal token embeddings through a projection operation. The patch embedding transformation for endogenous variables is formally defined by Equations (2) through (3):

$$\{s_1, s_2, s_N\} = \text{Patchify}(X), \quad (2)$$

$$P_{en} = \text{Embed}\{s_1, s_2, s_N\}. \quad (3)$$

where P denotes the patch length, $N = \frac{T}{P}$ represents the number of patches, and s_i indicates the i -th patch in the endogenous sequence P_{en} . The $\text{Embed}()$ function maps each patch of length P to a D -dimensional space through a trainable linear projection and positional embedding. Consequently, the patched endogenous variables $P_{en} \in \mathbb{R}^{N \times D}$ consist of N temporal tokens, each with D dimensions.

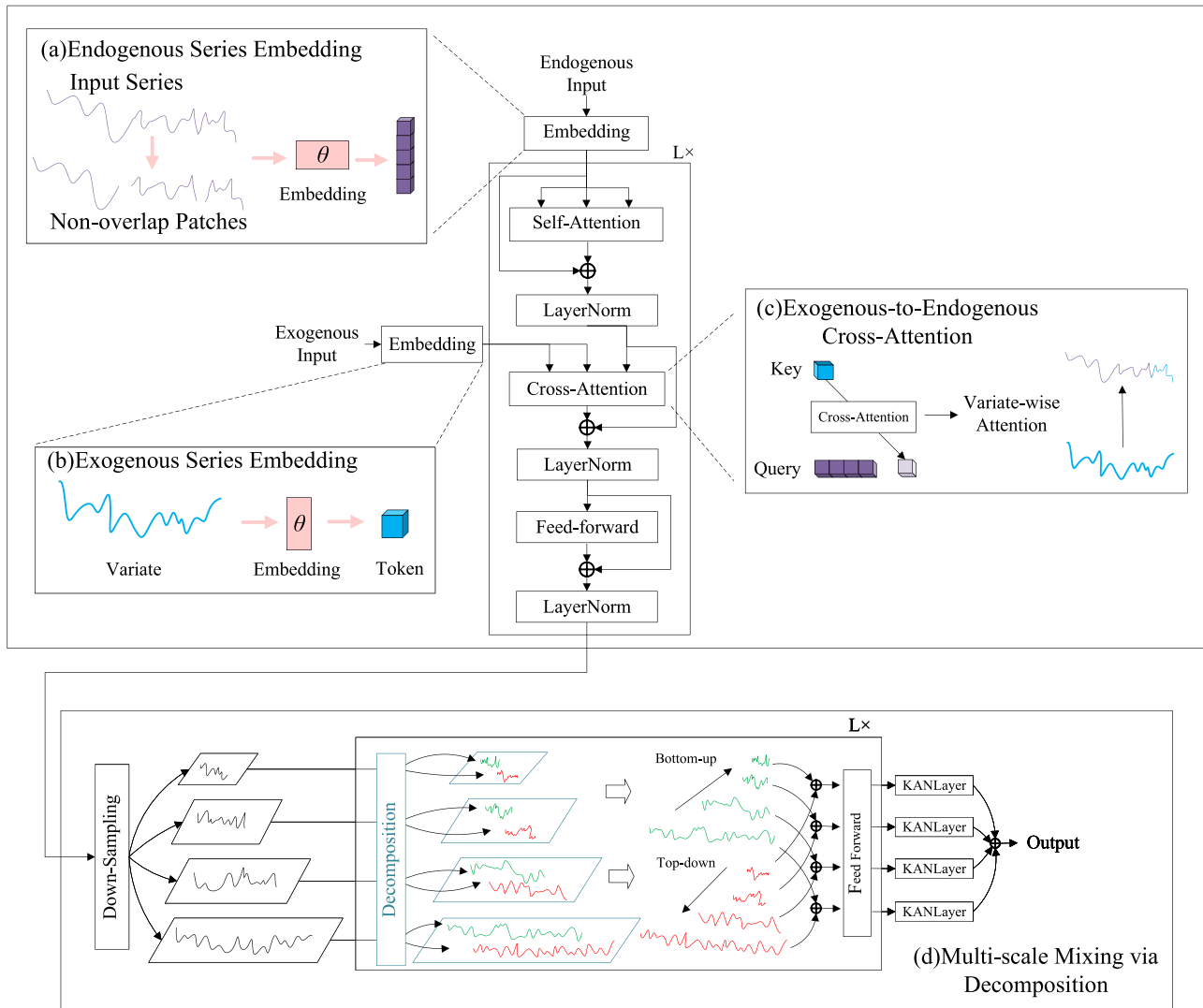


Fig. 2 The overall structure of TKANMiXer, (a) Input time series as endogenous variable tokens, (b) Last-column features as exogenous auxiliary variable tokens, (c) Fusion of endogenous and exogenous

variables via cross-attention mechanism, (d) Seasonal-trend decomposition with KAN-based prediction

The endogenous variables V_{en} and exogenous variables V_{ex} are formally derived from X and $Z_{1:T'}^{(i)}$ according to Equations (4) through (5):

$$V_{en} = EnVariateEmbed(X), \quad (4)$$

$$V_{ex,i} = ExVariateEmbed(Z_{1:T'}^{(i)}) \quad i \in (1, \dots, C). \quad (5)$$

The endogenous variables are projected into temporal tokens through Patch-wise Self-Attention, comprising multiple patch-wise temporal tokens $\hat{P}_{en}^l$ and variate-wise tokens $\hat{V}_{en}^l$, formulated as:

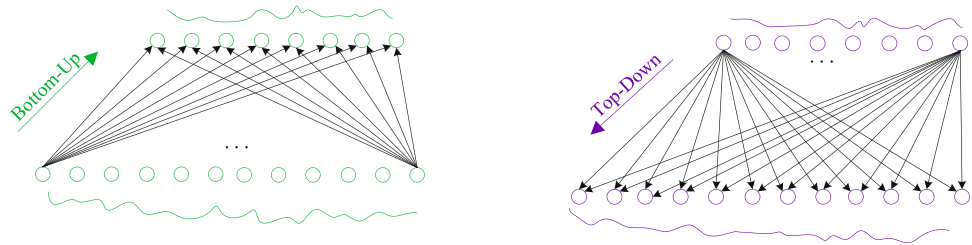
$$\hat{P}_{en}^l, \hat{V}_{en}^l = LN([P_{en}^l, V_{en}^l] + Self_Attention(P_{en}^l, V_{en}^l)), \quad (6)$$

where $l \in \{1, 2, \dots, M-1\}$ denotes the l -th patch. Further P_{en}^l and V_{en}^l represent the patch-wise and variate-wise tokens of the endogenous variables, respectively. $[\cdot, \cdot]$ denotes the concatenation of the patch-wise and series-wise tokens of the endogenous variable along the sequence dimension. $Self_Attention(\cdot)$ denotes the Multi-head Self-Attention over the endogenous tokens. $LN(\cdot)$ denotes Layer Normalization.

Using the cross-attention mechanism to integrate exogenous variables with endogenous variables, where the endogenous variables serve as queries and the exogenous variables serve as keys and values, establishing a connection between the two types of variables. The formula is as follows:

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Fig. 3 Seasonal-trend mixing architecture: left shows fine-to-coarse seasonal mixing, right demonstrates coarse-to-fine trend mixing



$$V_{en}^l = LN(\hat{V}_{en}^l + Cross_Attention(\hat{V}_{en}^l, V_{ex}, V_{ex})), \quad (7)$$

where the endogenous variable token $\hat{V}_{en}^l$, serving as the query, comes from the l -th block, while V_{ex} represents the exogenous variable tokens. The output of the cross-attention layer, V_{en}^l , can be used as the input for the next self-attention layer. By learning multivariate correlations from the exogenous variable tokens and propagating them to the endogenous temporal tokens, cross-modal information transfer between exogenous variable tokens and endogenous temporal tokens is achieved.

3.2 Multiscale mixing architecture

The time series is decomposed into multi-scale subsequences through downsampling, where fine-grained resolution captures local details while coarse-grained resolution characterizes global trends, thereby comprehensively revealing the complex dynamic relationships within the time series Wang et al. (2024). Specifically, the fused exogenous variable time series V_{en} is downsampled into multi-scale sequences $V_{en}^l \in \{v_{en_0}^l, v_{en_1}^l, \dots, v_{en_M}^l\}$, $v_{en_m}^l \in \mathbb{R}^{\lfloor \frac{P}{2^m} \rfloor \times d_{model}}$, $m \in \{0, \dots, M\}$ where P represents the observable length of V_{en} , M indicates the number of downsampling operations. The sequence $V_{en_0}^l = V_{en}^l$ serves as the finest-grained input. The multi-scale mixing formula is as follows:

$$V_{en}^l = PDM(V_{en}^{l-1}), l \in \{1, \dots, L\}, \quad (8)$$

where l denotes the number of layers, and $PDM()$ represents the Past-Decomposable-Mixing (PDM) blocks. The PDM captures multi-scale dependencies by decomposing time series data into seasonal and trend components, followed by separate feature extraction and cross-component mixing.

Fine-grained seasonal information enables more precise capture of high-frequency fluctuations and local temporal patterns. To effectively augment macro-level trends with detailed periodic variations, seasonal component mixing is implemented in a fine-to-coarse sequence direction. Following the series decomposition approach in Autoformer, the time series at each scale is decomposed into seasonal components $S^l \in \{s_0^l, s_1^l, \dots, s_M^l\}$ and trend components $T^l \in \{t_0^l, t_1^l, \dots, t_M^l\}$ where M represents the decomposition

level. Independent mixing of seasonal and trend components across multiple scales facilitates effective cross-scale information interaction. The complete mathematical formulation of the l -th PDM block is defined by Equations (9) through (10):

$$s_m^l, t_m^l = SeriesDecomp(v_{en_m}^l), m \in \{0, \dots, M-1\}, \quad (9)$$

$$v_{en_m}^l = v_{en_m}^{l-1} + FeedForward(S_Linear(s_m^l)_{m=0}^M + T_Linear(t_m^l)_{m=0}^M), \quad (10)$$

where $FeedForward(\cdot)$ consists of two linear layers with a GELU activation function in between, enabling channel-wise information interaction, while $S_Linear(\cdot) + T_Linear(\cdot)$ represents the mixing of seasonal and trend components.

The seasonal mixing process models subsequences in a fine-to-coarse hierarchical manner, augmenting macro-level variations with detailed seasonal patterns. The process operates on the set $S^l \in \{s_0^l, s_1^l, \dots, s_M^l\}$ across progressively coarser scales from 0 to M , as formalized below:

$$s_m^l = s_m^l + Bottom_Up_Linear(s_{m-1}^l), m \in \{1, \dots, M\}, \quad (11)$$

where $Bottom_Up_Linear(\cdot)$ consists of two linear layers with GELU activation in between for temporal information interaction along the time dimension. Its input dimension is $\frac{P}{2^{m-1}}$ and output dimension is $\frac{P}{2^m}$.

The trend mixing employs a coarse-to-fine hierarchical approach, which operates inversely to seasonal mixing, to simultaneously preserve long-term trend robustness and maintain short-term fluctuation sensitivity. Trend information is extracted from scale M to scale 0, as formalized below:

$$t_m^l = t_m^l + Top_Down_Linear(t_{m-1}^l), m \in \{1, \dots, M\}, \quad (12)$$

where $Top_Down_Linear(\cdot)$ consists of two linear layers with GELU activation in between for temporal information interaction along the time dimension. It transforms inputs from dimension $\frac{P}{2^{m-1}}$ to output dimension $\frac{P}{2^m}$.

The Periodic Decomposition Mixer (PDM) synergistically extracts seasonal-trend features via multi-scale hybrid modeling. The dual-path structure combines seasonal

Table 1 Dataset descriptions. Ex. and En. are abbreviations for the Exogenous variable and Endogenous variable, respectively

Tasks	Datasets	Dim	Ex.Descriptions	En. Descriptions	Series Length	Dataset Size	Information (Frequency)
Forecasting (Long-term)	ETTm1, ETTm2	7	Power Load Feature	Oil Temperature	96, 192, 336, 720	(34465, 11521, 11521)	15 mins
	ETTh1, ETTh2	7	Power Load Feature	Oil Temperature	96, 192, 336, 720	(8545, 2881, 2881)	15 mins
	Weather	21	Climate Feature	CO2-Concentration	96, 192, 336, 720	(36792, 5271, 10540)	10 mins
	Exchange	8	Exchange rate	Exchange rate	96, 192, 336, 720	(5120, 665, 1422)	Daily
	DJI	6	OHLCA Data	Trading volume	96, 192, 336	(2803, 334, 760)	Daily
	GSPC	6	OHLCA Data	Trading volume	96, 192, 336	(2803, 334, 760)	Daily
	HIS	6	OHLCA Data	Trading volume	96, 192, 336	(2735, 324, 741)	Daily
	FCHI	6	OHLCA Data	Trading volume	96, 192, 336	(2798, 332, 759)	Daily

The dataset size is organized in (Train, Validation, Test)

aggregation and trend fusion, enabling comprehensive multi-scale temporal representation (Fig. 3).

3.3 Multi-layer KAN

KANs decompose high-dimensional functions into simpler components but face computational bottlenecks from B-spline iterations. FastKAN is introduced, as a structural alternative to MLPs, utilizing radial basis functions (RBFs) for enhanced computational efficiency in approximation tasks. Gaussian RBFs (dependent on input-center distances) effectively substitute B-splines Li (2024), with N -center networks formulated as:

$$f(x) = \sum_{i=1}^N w_i \phi(|x - c_i|), \quad (13)$$

where w_i represents the trainable weights or coefficients of the RBFs. The Gaussian function, being one of the most widely used radial basis functions, computes its output based on the distance between the input x and center c_i , with its formulation as follows:

$$\phi(r) = \exp\left(-\frac{r^2}{2h^2}\right), \quad (14)$$

where r represents the radial distance, and h denotes the parameter controlling the function's width or spread.

The FastKAN is incorporated into the integrated framework to forecast multiscale time series processed through L PDM blocks, where the mixed hierarchical representations are denoted as $X^l \in \{x_0^l, x_1^l, \dots, x_M^l\}$, $x_m^l \in \mathbb{R}^{\lfloor \frac{P}{2^m} \rfloor \times d_{model}}$. The formula is as follows:

$$\hat{X}_m = KAN_m(X_m^L), m \in \{0, 1, \dots, M\}, \hat{X} = \sum_{m=0}^M \hat{X}_m, \quad (15)$$

where the $\hat{X}_m \in \mathbb{R}^{F \times C}$ represents the future prediction from the m -th scale sequence, and $KAN_m(\cdot)$ denotes the predictor for the m -th scale sequence. As an enhanced version of KAN, FastKAN maintains high prediction accuracy while effectively improving training speed, enabling broader applications of KAN. When employed as a predictor for multi-scale historical information, FastKAN demonstrably enhances the complementarity and accuracy of hybrid multi-scale forecasting.

4 Experiments

4.1 Datasets details

We conduct long-term forecasting experiments on 7 real-world datasets to evaluate the performance of our proposed TKANMiXer. Including: (1) ETT contains four subsets where ETTh1 and ETTh2 are hourly recorded, and ETTm1 and ETTm2 are recorded every 15 minutes. The endogenous variable is the oil temperature and the exogenous variables are 6 power load features. (2) Weather records 21 meteorological factors collected every 10 minutes from the Weather Station of the Max Planck Biogeochemistry Institute in 2020. In our experiment, we use the Wet Bulb factor as the endogenous variable to be predicted and the other indicators as exogenous variables. (3) Exchange recorded the exchange rate of the currency against the US dollar from 1990 to 2010. (4) Four major stock indices from different regions are selected for evaluation: CAC40 (FCHI) from France, representing the 40 largest companies on the Paris Stock Exchange; Hang Seng (HSI) from Hong Kong, representing 50 major companies listed on the Hong Kong Stock Exchange; S&P 500 (GSPC) from the United States, covering 500 large U.S. companies; and Dow Jones Industrial Average (DJI) from the U.S., made up of 30 major industrial companies. These indices offer a diverse view of global market performance. The dataset spans from January 1, 2007, to December 29, 2023. We treat the trading volume (V) as an endogenous variable, while considering the opening price

Table 2 Long-term forecasting results. Avg means the average results from all four prediction lengths

MODELS METRIC	TKAN- Mixer (OURS)	TimeXer (2024)		TimeMixer (2024)		ITrans- former (2024)		TimesNet (2023)		DLinear (2023)		PatchTST (2024)		TiDE (2023)		FEDformer (2022b)		Autoformer (2021)		Crossformer (2023)		SCINet (2022a)			
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE		
Exchange	96	0.083	0.203	0.086	0.205	<i>0.085</i>	<i>0.204</i>	0.086	0.206	0.107	0.234	0.088	0.218	0.088	0.205	0.094	0.218	0.148	0.278	0.197	0.323	0.256	0.367	0.267	0.396
	192	0.175	0.298	0.182	0.302	0.190	0.310	0.177	0.299	0.226	0.344	0.176	0.315	<i>0.176</i>	<i>0.299</i>	0.184	0.307	0.271	0.315	0.300	0.369	0.470	0.509	0.351	0.459
	336	0.297	0.397	0.367	0.438	0.356	0.433	0.331	<i>0.417</i>	0.367	0.448	0.313	0.427	0.301	0.397	0.349	0.431	0.460	0.427	0.509	0.524	1.268	0.883	1.324	0.853
	720	0.802	0.677	0.887	0.710	0.927	0.714	0.847	<i>0.691</i>	0.964	0.746	0.839	0.695	0.901	0.714	0.852	0.698	1.195	0.695	1.447	0.941	1.767	1.068	1.058	0.797
	AVG	0.339	0.394	0.381	0.414	0.390	0.415	0.360	<i>0.403</i>	0.416	0.443	0.402	0.397	0.412	0.367	0.404	0.370	0.413	0.519	0.429	0.613	0.539	0.940	0.707	0.750
ETTh1	96	0.380	<i>0.404</i>	0.385	<i>0.404</i>	0.385	0.402	0.386	0.405	<i>0.384</i>	0.402	0.397	0.412	0.460	0.447	0.479	0.464	0.395	0.424	0.449	0.459	0.423	0.448	0.654	0.599
	192	<i>0.436</i>	0.433	0.433	0.438	0.443	<i>0.430</i>	0.441	0.436	<i>0.436</i>	0.429	0.446	0.441	0.512	0.447	0.525	0.492	0.469	0.470	0.500	0.482	0.471	0.474	0.719	0.631
	336	0.460	0.446	<i>0.469</i>	<i>0.450</i>	0.512	0.470	0.487	0.458	0.638	0.469	0.489	0.467	0.546	0.496	0.565	0.515	0.490	0.477	0.521	0.496	0.570	0.546	0.778	0.659
	720	0.470	0.468	0.504	0.495	0.497	0.476	0.503	<i>0.491</i>	0.521	0.500	0.513	0.510	0.554	0.517	0.594	0.558	0.598	0.544	0.514	0.512	0.653	0.621	0.836	0.699
	AVG	0.437	0.438	<i>0.448</i>	<i>0.447</i>	0.459	<i>0.444</i>	0.454	0.447	0.495	0.450	0.461	0.457	0.516	0.484	0.541	0.507	0.498	0.484	0.496	0.487	0.529	0.522	0.747	0.647
ETTh2	96	0.288	0.339	0.286	0.338	0.289	0.342	0.297	0.349	0.340	0.374	0.340	0.394	0.308	0.355	0.400	0.440	0.358	0.397	0.346	0.388	0.745	0.584	0.707	0.621
	192	0.372	0.397	0.364	0.390	0.378	0.397	0.380	0.400	0.402	0.414	0.482	0.479	0.393	0.405	0.528	0.509	0.429	0.439	0.456	0.452	0.877	0.656	0.860	0.689
	336	0.421	0.432	0.415	0.423	0.432	0.434	0.428	<i>0.432</i>	0.452	0.452	0.591	0.541	0.427	0.436	0.643	0.571	0.496	0.487	0.482	0.486	1.043	0.731	1.000	0.744
	720	0.431	0.444	0.444	0.451	0.464	0.464	0.427	<i>0.445</i>	0.462	0.468	0.839	0.661	0.436	0.450	0.874	0.679	0.463	0.474	0.515	0.511	1.104	0.763	1.249	0.838
	AVG	0.378	0.403	0.377	0.401	0.390	0.409	0.383	0.407	0.414	0.427	0.563	0.519	0.391	0.411	0.611	0.550	0.437	0.449	0.450	0.459	0.942	0.684	0.954	0.723
ETTm1	96	0.314	0.353	0.322	0.358	0.317	0.356	0.334	0.368	0.338	0.375	0.346	0.374	0.352	0.374	0.364	0.387	0.379	0.419	0.505	0.475	0.404	0.426	0.418	0.438
	192	0.359	0.379	0.362	0.384	0.367	0.384	0.377	0.391	0.374	0.387	0.382	0.391	0.390	0.393	0.398	0.404	0.426	0.441	0.553	0.496	0.450	0.451	0.439	0.450
	336	0.393	0.406	0.395	<i>0.407</i>	0.391	0.406	0.426	0.420	0.410	0.411	0.415	0.415	0.421	0.414	0.428	0.425	0.445	0.459	0.621	0.537	0.532	0.515	0.490	0.485
	720	0.450	<i>0.443</i>	<i>0.451</i>	0.441	0.454	0.441	0.491	0.459	0.478	0.450	0.473	0.451	0.462	0.449	0.487	0.461	0.543	0.490	0.671	0.561	0.666	0.589	0.595	0.550
	AVG	0.379	0.395	0.383	0.398	0.382	0.397	0.407	0.410	0.400	0.406	0.404	0.408	0.406	0.407	0.419	0.419	0.448	0.452	0.588	0.517	0.513	0.496	0.485	0.481
ETTm2	96	0.172	0.256	0.170	0.256	0.175	0.257	0.180	0.264	0.187	0.267	0.193	0.293	0.183	0.270	0.207	0.305	0.203	0.287	0.225	0.339	0.287	0.366	0.286	0.377
	192	0.239	<i>0.301</i>	0.237	0.299	0.240	0.302	0.250	0.309	0.249	0.309	0.284	0.361	0.255	0.314	0.290	0.364	0.269	0.328	0.281	0.340	0.414	0.492	0.399	0.445
	336	0.302	0.341	0.302	0.342	0.303	0.343	0.311	0.348	0.321	0.351	0.382	0.429	0.309	0.347	0.377	0.422	0.325	0.366	0.339	0.372	0.597	0.542	0.637	0.591
	720	0.399	0.398	0.391	0.394	0.392	0.396	0.412	0.407	0.408	0.403	0.558	0.525	0.412	0.404	0.558	0.524	0.421	0.415	0.443	0.432	0.730	1.042	0.960	0.735
	AVG	0.278	<i>0.324</i>	0.275	0.323	0.277	<i>0.324</i>	0.288	0.332	0.291	0.333	0.354	0.402	0.290	0.334	0.358	0.404	0.305	0.349	0.327	0.371	0.757	0.610	0.571	0.537
weather	96	0.179	0.225	0.178	0.226	0.163	0.209	<i>0.174</i>	<i>0.214</i>	0.172	0.220	0.195	0.252	0.186	0.227	0.202	0.261	0.217	0.296	0.266	0.336	0.195	0.271	0.221	0.306
	192	0.234	0.274	0.230	0.272	<i>0.211</i>	0.254	0.221	0.254	0.219	0.261	0.237	0.295	0.234	0.265	0.242	0.298	0.276	0.336	0.307	0.367	0.209	0.277	0.261	0.340
	336	0.293	0.317	0.291	0.315	0.263	0.293	0.278	0.296	0.280	0.306	0.282	0.331	0.284	0.301	0.287	0.335	0.339	0.380	0.359	0.395	0.273	0.332	0.309	0.378
	720	0.385	0.375	0.378	0.369	0.344	0.348	0.358	0.347	0.365	0.359	0.345	0.382	0.356	0.349	0.351	0.386	0.403	0.428	0.419	0.428	0.379	0.401	0.377	0.427
	AVG	0.273	0.298	0.269	0.296	0.245	0.276	0.258	0.278	0.259	0.287	0.265	0.315	0.265	0.285	0.271	0.320	0.309	0.360	0.338	0.382	0.264	0.320	0.292	0.363

The best results are highlighted in red and the second-best are in blue

Table 3 A comparison of multivariate long-term forecast results in a financial time series

MODELS METRIC		TKANMiXer (OURS)		TimeXer (2024)		TimeMixer (2024)		ITransformer (2024)		TimesNet (2023A)		DLinear (2023)		PatchTST (2024)		TiDE (2023)		FEDformer (2022b)		Autoformer (2021)		Crossformer (2023)		SCINet (2022a)	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
DJI	96	0.208	0.327	0.209	0.329	0.238	0.359	0.289	0.407	0.242	0.360	0.276	0.397	0.214	0.336	0.255	0.373	0.304	0.431	0.332	0.453	2.064	1.332	0.231	0.345
	192	0.331	0.434	0.348	0.449	0.360	0.456	0.416	0.497	0.400	0.486	0.481	0.542	0.379	0.478	0.431	0.506	0.490	0.561	0.506	0.568	2.381	1.458	0.346	0.445
	336	0.499	0.556	0.514	0.567	0.574	0.587	0.671	0.651	0.574	0.606	0.749	0.682	0.503	0.554	0.562	0.599	0.682	0.686	0.676	0.666	8.155	2.641	0.535	0.568
	AVG	0.346	0.439	0.357	0.448	0.391	0.467	0.459	0.518	0.405	0.484	0.502	0.540	0.365	0.456	0.416	0.493	0.492	0.559	0.505	0.562	4.200	1.810	0.371	0.453
GSPC	96	0.287	0.388	0.288	0.395	0.348	0.460	0.455	0.549	0.328	0.434	0.429	0.530	0.318	0.430	0.418	0.519	0.492	0.580	0.521	0.603	4.990	2.063	0.328	0.426
	192	0.525	0.541	0.521	0.549	0.607	0.623	0.682	0.673	0.615	0.610	0.775	0.722	0.569	0.570	0.707	0.687	0.804	0.748	0.821	0.755	5.120	2.094	0.553	0.557
	336	0.929	0.782	0.945	0.768	1.291	0.855	1.148	0.884	1.025	0.831	1.223	0.881	0.938	0.785	0.993	0.822	1.166	0.911	1.177	0.908	11.388	3.106	0.952	0.761
	AVG	0.580	0.570	0.585	0.571	0.749	0.646	0.762	0.702	0.656	0.625	0.809	0.711	0.608	0.595	0.706	0.676	0.821	0.746	0.840	0.755	7.166	2.421	0.611	0.581
HSI	96	0.527	0.534	0.535	0.538	0.602	0.594	0.593	0.570	0.567	0.570	0.615	0.605	0.574	0.573	0.595	0.559	0.718	0.689	0.706	0.658	0.580	0.592	0.569	0.550
	192	0.812	0.704	0.817	0.712	0.881	0.747	0.930	0.764	0.880	0.746	0.790	0.701	0.839	0.719	0.921	0.759	1.046	0.824	1.055	0.813	0.682	0.636	0.880	0.741
	336	1.246	0.864	1.472	0.969	1.935	1.087	1.631	1.032	1.512	0.991	1.057	0.838	1.614	1.020	1.496	0.967	1.924	1.143	1.906	1.112	1.363	0.929	1.612	0.993
	AVG	0.862	0.701	0.941	0.740	1.139	0.809	1.051	0.789	0.986	0.769	0.821	0.715	1.009	0.771	1.004	0.762	1.229	0.885	1.222	0.861	0.875	0.719	1.020	0.761
FCHI	96	0.321	0.436	0.327	0.439	0.417	0.503	0.368	0.473	0.354	0.458	0.708	0.687	0.395	0.491	0.371	0.476	0.523	0.595	0.653	0.658	1.836	1.177	0.429	0.494
	192	0.740	0.681	0.768	0.682	0.875	0.719	0.845	0.734	0.944	0.767	1.792	1.100	1.199	0.781	0.809	0.701	0.990	0.819	1.010	0.825	2.389	1.375	0.807	0.712
	336	1.168	0.831	1.331	0.875	1.859	0.961	1.295	0.878	1.469	0.935	2.750	1.419	1.271	0.849	1.272	0.866	1.417	0.927	1.374	0.909	5.679	2.171	1.407	0.903
	AVG	0.743	0.649	0.809	0.665	1.050	0.728	0.836	0.695	0.922	0.720	1.750	1.069	0.955	0.707	0.817	0.681	0.977	0.780	1.012	0.797	3.301	1.574	0.881	0.703

Avg means the average results from all four prediction lengths. The best results are highlighted in red and the second-best are in blue

Table 4 Performance Comparison: Endogenous-Only (En) vs. Endogenous+Exogenous (En+Ex)

Datasets Metric		Exchange		ETTh1		ETTh2		ETTm1		ETTm2	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
En	96	0.088	0.210	0.384	0.398	0.296	0.345	0.321	0.358	0.185	0.273
	192	0.172	0.303	0.462	0.450	0.397	0.408	0.366	0.386	0.254	0.318
	336	0.330	0.419	0.487	0.456	0.425	0.435	0.385	0.400	0.316	0.357
	720	0.843	0.693	0.514	0.496	0.449	0.457	0.451	0.442	0.414	0.414
	AVG	0.358	0.406	0.462	0.450	0.392	0.411	0.381	0.397	0.292	0.341
En+Ex	96	0.083	0.203	0.382	0.404	0.288	0.339	0.315	0.353	0.174	0.256
	192	0.175	0.298	0.436	0.433	0.372	0.397	0.359	0.379	0.239	0.301
	336	0.297	0.397	0.460	0.446	0.421	0.432	0.393	0.406	0.302	0.341
	720	0.802	0.677	0.470	0.468	0.431	0.444	0.450	0.443	0.399	0.398
	AVG	0.339	0.394	0.437	0.438	0.378	0.403	0.379	0.395	0.279	0.324

Table 5 Ablation study on decomposition, seasonal mixing ($\nearrow$), and trend mixing ($\swarrow$) components in ETTh1 & ETTm1 datasets

Case	Decompose	Mixing		ETTh1		ETTm1	
		Seasonal	Trend	MSE	MAE	MSE	MAE
①	✓	$\nearrow$	$\swarrow$	0.460	0.446	0.393	0.406
②	✓	×	$\swarrow$	0.464	0.449	0.394	0.409
③	✓	$\nearrow$	×	0.470	0.451	0.396	0.409
④	×	$\nearrow$		0.463	0.448	0.395	0.407
⑤	×	$\swarrow$		0.464	0.449	0.395	0.405
⑥	✓	×		0.471	0.451	0.397	0.410
⑦	×	×		0.476	0.455	0.400	0.411

Notation: with component (✓), without component (×). Fixed settings: look-back length = 96, prediction length = 336

Table 6 Multi-layer KAN Ablation Study

Datasets Metric		Exchange		ETTh1		ETTh2		ETTm1		ETTm2	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
MLP		0.350	0.427	0.485	0.456	0.427	0.438	0.403	0.413	0.304	0.344
Multi layer KAN		0.083	0.203	0.382	0.404	0.288	0.339	0.315	0.353	0.174	0.256

Table 7 The MSE and MAE results with different numbers of scales (M) on Exchange and other datasets

Number of scales	Exchange		ETTh1		ETTh2		ETTm1		ETTm2	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
1	0.083	0.202	0.389	0.408	0.299	0.351	0.321	0.359	0.173	0.259
2	0.085	0.204	0.389	0.406	0.292	0.344	0.315	0.355	0.173	0.257
3	0.085	0.204	0.388	0.407	0.309	0.360	0.322	0.359	0.172	0.256
4	0.084	0.203	0.388	0.412	0.298	0.345	0.320	0.356	0.172	0.256

(O), high price (H), low price (L), closing price (C), and adjusted close (A) as exogenous variables. The complete dataset specifications are provided in Table 1.

4.2 Baseline

We conducted comprehensive comparisons between TKANMiXer and SOTA long-term forecasting models, including iTransformer Liu et al. (2023), TimesNet Wu et al. (2022), DLinear Zeng et al. (2023), PatchTST Nie et al. (2022), TiDE Das et al. (2023), FEDformer Zhou et al. (2022), Autoformer Wu et al. (2021), Crossformer Zhang and Yan (2023), SCINet Liu et al. (2022).

4.3 Experimental setup

To ensure fair comparisons, we adopt the same look-back window length $T = 96$ and the same prediction length $F = \{96, 192, 336, 720\}$. We utilize the L2 loss for model training and use Mean Square Error (MSE) and Mean Absolute Error (MAE) metrics to evaluate the performance of each method.

All the experiments are implemented in PyTorch and conducted on an NVIDIA A100-SXM4-80GB GPU and Intel Xeon Platinum 8168 CPU with 16 cores. We utilize the L2 loss for model training. The number of scales M is set

Table 8 A comparison of model parameters (Params), multiply-accumulate operations (MACs) and GPU memory for TKANMiXer and three other models

Datasets	Exchange			ETTh1			ETTh2			ETTm1			ETTm2		
	Params	MACs	Memory	Params	MACs	Memory	Params	MACs	Memory	Params	MACs	Memory	Params	MACs	Memory
TKANMiXer	0.62M	0.11G	583.98MB	0.08M	0.01G	80.09MB	0.16M	0.02G	262.48MB	0.08M	0.01G	94.40MB	1.00M	0.68G	1365.43MB
TimeXer	1.78M	0.08G	117.73MB	0.75M	0.02G	35.49MB	1.26M	0.04G	303.31MB	1.78M	0.07G	554.76MB	1.78M	0.07G	554.76MB
Autoformer	10.54M	1.19G	1694.74MB	10.54M	1.19G	1694.57MB	10.54M	1.19G	1694.57MB	10.54M	1.19G	1694.57MB	10.54M	1.19G	1118.56MB
PatchTST	6.90M	0.61G	372.92MB	3.75M	0.27G	210.49MB	10.06M	0.80G	477.25MB	3.75M	0.27G	210.49MB	10.06M	0.80G	488.18MB

To ensure a fair comparison, we fix the prediction length $F = 96$ and the input length $T = 96$, and set the input batch size to 32

according to the time series length to trade off performance and efficiency.

4.4 Metric details

Regarding metrics, we utilize the mean square error (MSE) and mean absolute error (MAE) for long-term forecasting. The calculations of these metrics are:

$$MSE = \left(\sum_{i=1}^F (X_i - \hat{X}_i)^2 \right)^{\frac{1}{2}}, \quad (16)$$

$$MAE = \sum_{i=1}^F |X_i - \hat{X}_i|, \quad (17)$$

where $X_i, \hat{X}_i \in \mathbb{R}^{F \times C}$ are the ground truth and prediction results of the future with F time pints and C dimensions. X_i means the i -th future time point.

4.5 Experimental results

The forecasting results (Table 2) highlight TKANMiXer's superior performance, with best and second-best results marked in bold red and blue respectively. Experimental results show that TKANMiXer exhibited superior forecasting performance on most datasets. In addition, TKANMiXer is consistent with TimeXer and TimeMixer in long-term prediction, demonstrating the feasibility of a well-designed variable fusion and sequence decomposition architecture for accurate prediction. Compared with other state-of-the-art methods, TKANMiXer introduces a fusion-decomposition-learning-hybrid framework of endogenous and exogenous variables, which integrates the characteristics of multi-order KAN with a layer-mixing framework to achieve advanced performance in a wide range of long-term prediction tasks.

Additionally, we also verified on the financial time series dataset, with the forecasting results presented in Table 3. Due to dataset length constraints, the prediction horizons for the four stock indices (FCHI, HSI, GSPC, and DJI) were set to $F = \{96, 192, 336\}$. Evaluation across these four geographically diverse market indices demonstrates that TKANMiXer achieves SOTA performance in long-term forecasting of financial time series data.

5 Ablation study

This chapter will focus on three critical components: (1) endogenous and exogenous variables, (2) the combination of seasonal trends, (3) the hybrid KAN for multi-prediction.

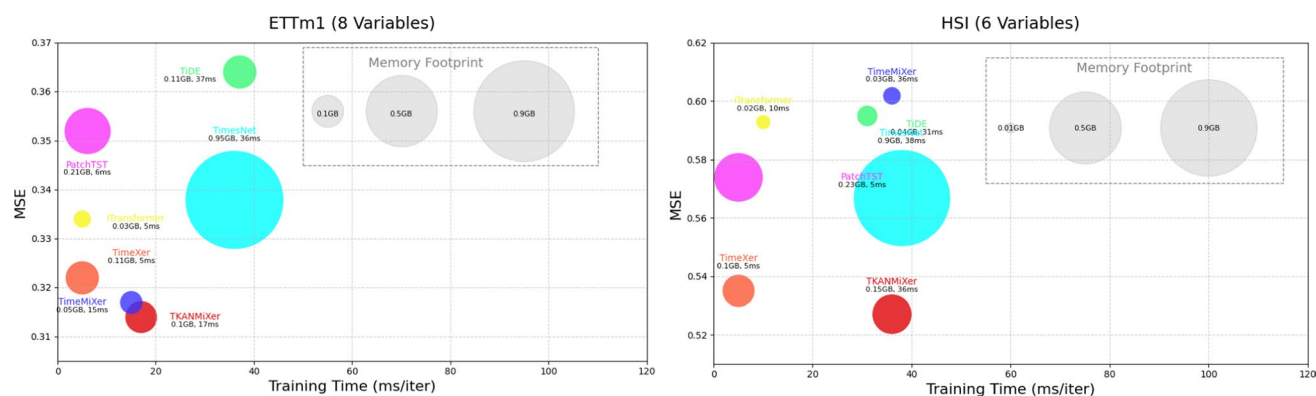


Fig. 4 Model efficiency comparison on ETTm1 and HSI datasets

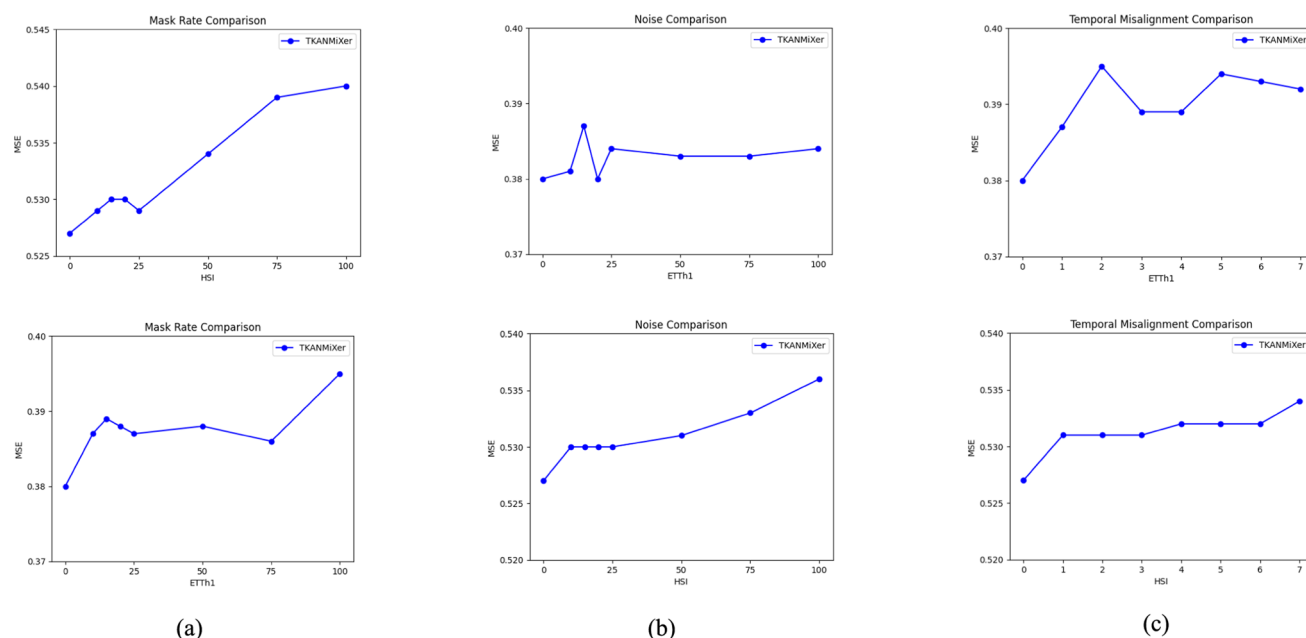


Fig. 5 Robustness Evaluation on ETTh1 and HSI Datasets with a Focus on (a) Masking Rate, (b) Noise, and (c) Temporal Misalignment

5.1 Exogenous integration analysis

Table 4 compares endogenous-only (En) vs. endogenous + exogenous (En+Ex) predictions, with best results bolded. En+Ex consistently outperforms En-only, validating exogenous integration in TKANMiXer. Experiments employed a fixed input length of 96 time steps and evaluated output lengths 96, 192, 336 and 720 steps, with averaged results reported across all output length.

5.2 Seasonal and trend mixing

The seasonal-trend decomposition hybrid framework effectively handles complex time-series variations when integrating exogenous variables. As shown in Table 5, TKANMiXer employs this approach with fine-grained mixing for seasonal components and coarse-grained mixing for trends. This dual

decomposition strategy successfully captures intricate temporal patterns, demonstrating the critical role of the decomposition mechanism in model performance.

5.3 Multi-layer KAN prediction

To assess KAN's efficacy in TKANMiXer, we compared two variants: (1) an MLP baseline with all KANs replaced by multilayer perceptrons; (2) a multi-layer KAN for hierarchical temporal feature extraction. Results (Table 6) show multi-layer KAN outperforms others across all metrics, demonstrating its superior multi-scale pattern modeling via hierarchical feature extraction as a robust MLP alternative.

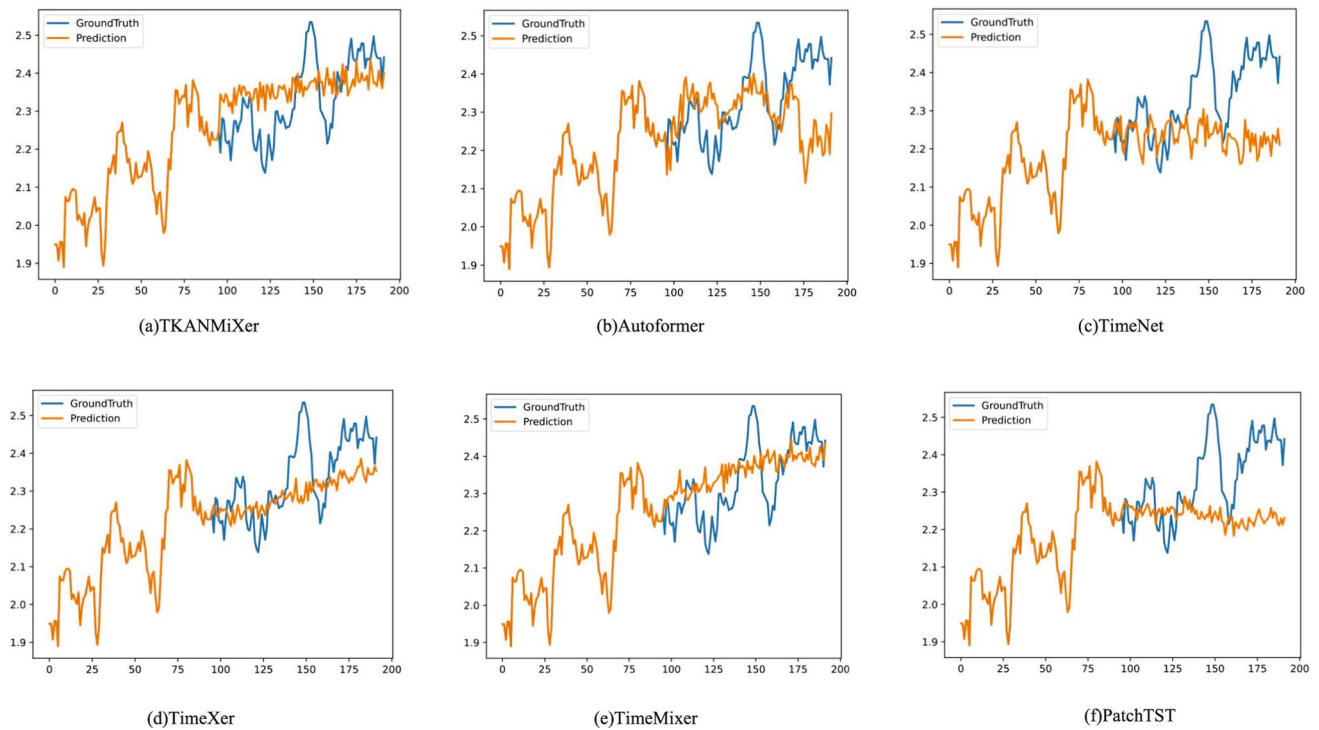


Fig. 6 Prediction cases from Exchange by different models under the input-96-predict-96 settings. Blue lines are the ground truths and orange lines are the model predictions

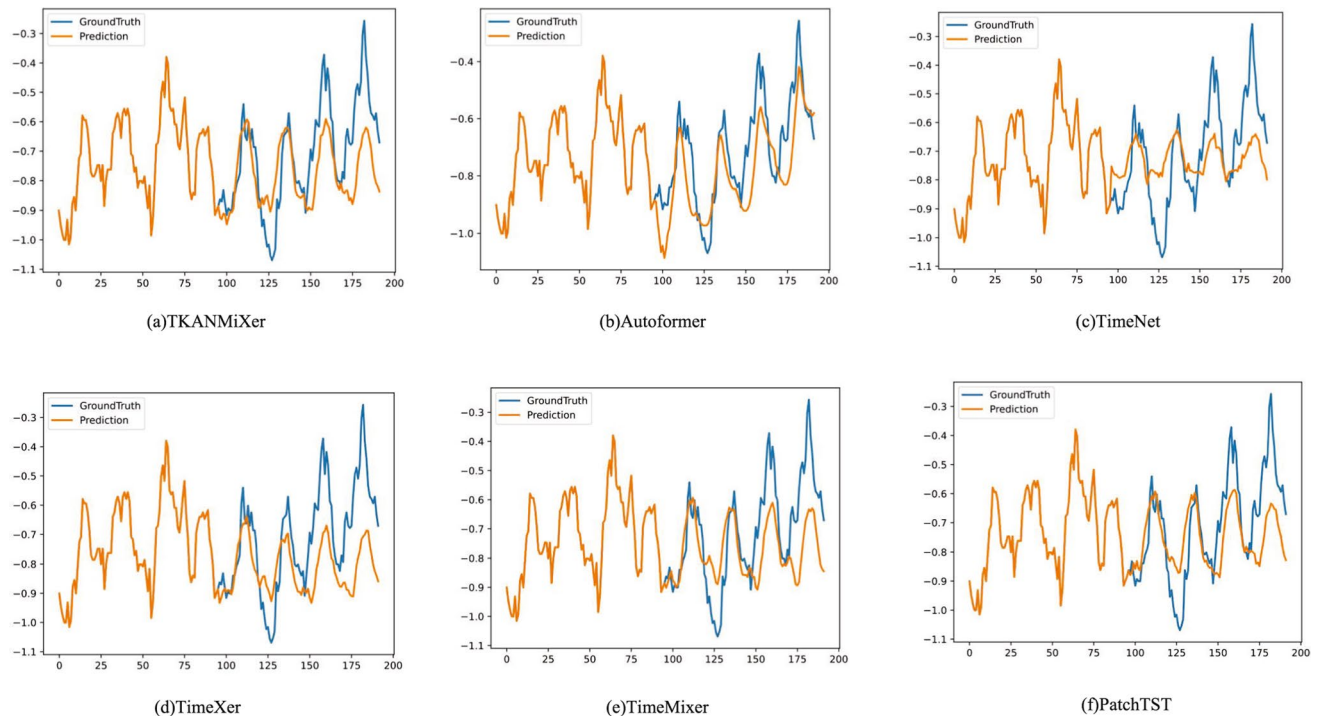


Fig. 7 Prediction cases from ETTh1 by different models under input-96-predict-96 settings

5.4 Impact of mixed-scale parameters

We evaluate the impact of the number of scales on model

prediction performance. As shown in Table 7, the influence of different numbers of scales on performance demonstrates

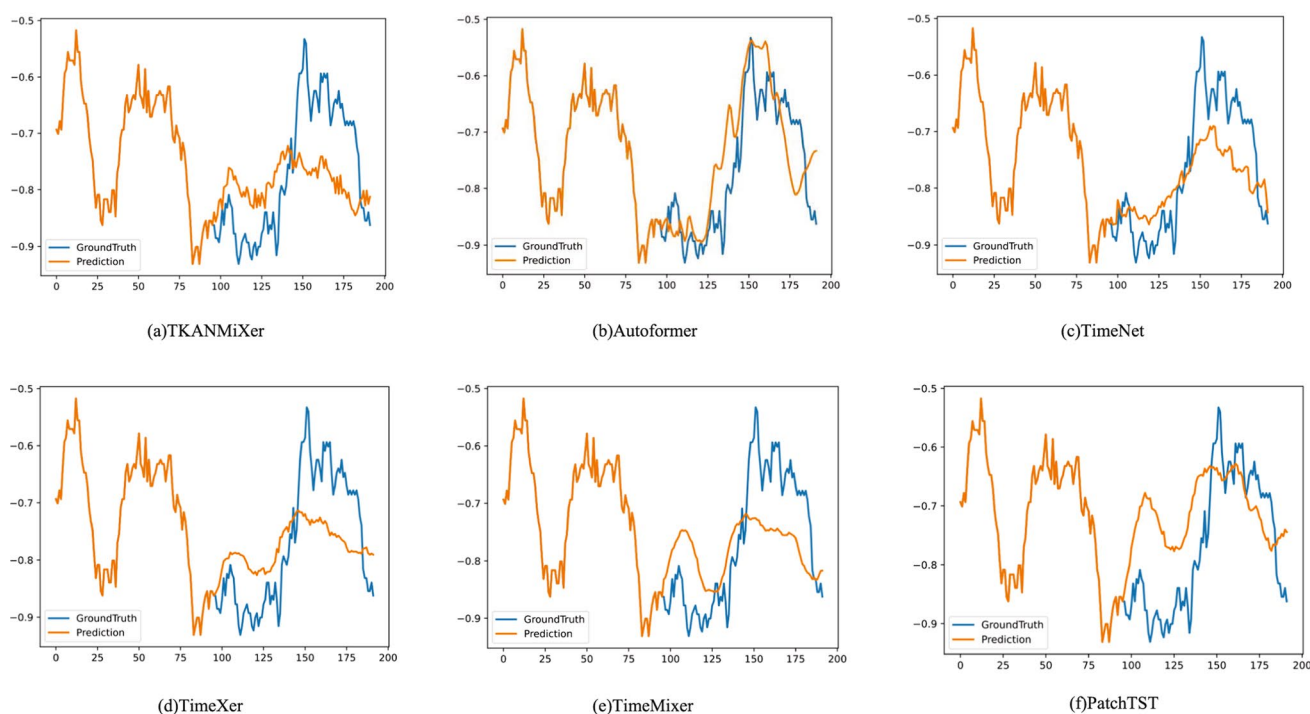


Fig. 8 Prediction cases from ETTm1 by different models under input-96-predict-96 settings

that the model generally achieves optimal performance when the number of scales is 2 or more.

5.5 Model efficiency

In KANs, when the network depth L and width N are the same, the number of parameters in KAN is higher compared to MLPs. However, KANs require significantly fewer N (width) than MLPs, thereby achieving parameter efficiency Liu et al. (2024). To demonstrate that TKANMiXer is an efficient and lightweight framework, we compared it with three other models in terms of parameters (Params), multiply-accumulate operations (MACs), and GPU memory usage. The results are shown in Table 8.

To comprehensively evaluate the efficiency of TKANMiXer, we compared its training speed and memory usage against six baseline models on both the ETTm1 and HSI datasets. The analysis reveals that although TKANMiXer achieves optimal forecasting accuracy, it does not demonstrate a significant advantage in computational efficiency. This outcome aligns with expectations, as the enhanced representational capacity and approximation capabilities of the KAN operator, compared to standard MLPs, inevitably introduce higher computational overhead. The principal contribution of this work lies in demonstrating the feasibility of achieving substantial performance gains at a moderate computational cost, thereby offering a valuable new

solution for scenarios with stringent accuracy requirements. The model comparison is presented in Figure 4.

High-performance computing not only enhances quality of service but also maintains energy awareness Ali et al. (2017). Compared to other models, TKANMiXer utilizes smaller hyperparameter values, thereby achieving significant advantages in both parameter count (Params) and computational complexity (MACs) while maintaining efficient GPU memory utilization. This combination of efficiency makes it particularly suitable for long-term forecasting tasks, where it exhibits SOTA performance.

5.6 Robustness test

In real-world applications, exogenous variables often suffer from quality degradation due to issues such as data missing, noise interference, and temporal misalignment. To simulate these scenarios, we systematically construct exogenous variable inputs with varying quality levels through random masking, additive Gaussian noise, and artificial temporal misalignment. While maintaining the endogenous variables as complete high-quality historical data, we progressively degrade the quality of exogenous variables to evaluate the model's robustness to noisy inputs. Experimental results demonstrate that as the quality of exogenous sequences decreases, the model's forecasting performance shows corresponding degradation. However, TKANMiXer maintains competitive prediction accuracy even with mildly degraded

exogenous variables, indicating its adaptability to low-quality data scenarios. Comparative results are shown in Figure 5.

5.7 Showcases

To evaluate model performance, we compare the forecasting results of TKANMiXer with baseline models on the Exchange, ETTh1, and ETTm1 test datasets, as visually presented in Figures 6, 7, and 8. The time series predictions demonstrate that TKANMiXer more accurately captures detailed temporal variations and complex patterns, achieving superior forecasting precision. These results validate the effectiveness of integrating KAN-based variable fusion with multi-scale seasonal-trend decomposition.

6 Conclusion

We propose TKANMiXer, a novel KAN-based architecture that synergistically combines cross-variable fusion and multi-scale seasonal-trend decomposition through a hierarchical framework. The model's hybrid KAN layers enable progressive extraction of multi-resolution temporal features, simultaneously capturing both local periodic patterns and global cross-cycle dependencies. Through its unique combination of attention-based variable interaction and adaptive scale decomposition, TKANMiXer effectively addresses the key challenges of long-term temporal dependencies and multi-scale feature integration in time series forecasting. Extensive experiments demonstrate that TKANMiXer achieves state-of-the-art forecasting accuracy across multiple benchmarks while maintaining competitive computational efficiency compared to existing approaches.

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Data Availability The data from this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

Ethics declarations This article does not contain any studies with human participants and/or animals performed by any of the authors.

Ethical and informed consent for data used Authors have cited any publicly available data on which the conclusions of the paper rely.

References

- Abueidda DW, Pantidis P, Mobasher ME (2025) Deepokan: Deep operator network based on kolmogorov arnold networks for mechanics problems. *Comput Methods Appl Mech Eng* 436:117699
- Ali R, Ali A, Naeem HMY, Asad M, Alsarhan T, Heyat BB (2024) A comprehensive survey of deep learning-based traffic flow prediction models for intelligent transportation systems. *ICCK Transactions on Advanced Computing and Systems* 1(3):117–137
- Ali R, Li Q, Chen J, Huang D, Lu X (2021) The spatial characteristics of hourly rainfall induced by tropical cyclones along the south china coast. *Earth and Space Science* 8(11):2020–001623
- Ali R, Shen Y, Huang X, Zhang J, Ali A (2017) Vmr: virtual machine replacement algorithm for qos and energy-awareness in cloud data centers. In: 2017 IEEE International Conference on Computational Science and Engineering (CSE) and IEEE International Conference on Embedded and Ubiquitous Computing (EUC), vol. 2, pp. 230–233. IEEE
- Bozorgasl Z, Chen H Wav-kan: Wavelet kolmogorov-arnold networks, 2024. arXiv preprint [arXiv:2405.12832](https://arxiv.org/abs/2405.12832)
- Challu C, Olivares KG, Oreshkin BN, Ramirez FG, Canseco MM, Dubrawski A (2023) Nhits: Neural hierarchical interpolation for time series forecasting. In: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, pp. 6989–6997
- Cleveland RB, Cleveland WS, McRae JE, Terpenning I (1990) Stl: A seasonal-trend decomposition. *J off Stat* 6(1):3–73
- Danish MU, Grolinger K (2025) Kolmogorov-arnold recurrent network for short term load forecasting across diverse consumers. *Energy Rep* 13:713–727
- Das A, Kong W, Leach A, Mathur, S, Sen R, Yu R (2023) Long-term forecasting with tide: Time-series dense encoder. arXiv preprint [arXiv:2304.08424](https://arxiv.org/abs/2304.08424)
- Genet R, Inzirillo H (2024) Tkan: Temporal kolmogorov-arnold networks. arXiv preprint [arXiv:2405.07344](https://arxiv.org/abs/2405.07344)
- Genet R, Inzirillo H (2024) A temporal kolmogorov-arnold transformer for time series forecasting. ArXiv
- Jamadar IS, Kumar K, Jadhav P, Bhalerao P, Afgan S Prediction of infiltrating ductal carcinoma using morlet wavelet integrated kolmogorov arnold network
- Livieris IE (2024) C-kan: A new approach for integrating convolutional layers with kolmogorov-arnold networks for time-series forecasting. *Mathematics* 12(19):3022
- Li Z (2024) Kolmogorov-arnold networks are radial basis function networks. arXiv preprint [arXiv:2405.06721](https://arxiv.org/abs/2405.06721)
- Li Q, Ali R, Zhang J, He L, Wu Z, Ye Y, Zhang L, Chan P-W (2025) Spatial characteristics and prediction of tropical cyclone-induced storm surge along the guangdong and hong kong coast. *Atmos Res* 320:108052
- Liu Y, Hu T, Zhang H, Wu H, Wang S, Ma L, Long M (2023) itransformer: Inverted transformers are effective for time series forecasting. arXiv preprint [arXiv:2310.06625](https://arxiv.org/abs/2310.06625)

- Liu Z, Wang Y, Vaidya S, Ruehle F, Halverson J, Soljačić M, Hou TY, Tegmark M (2024) Kan: Kolmogorov-arnold networks. arXiv preprint [arXiv:2404.19756](https://arxiv.org/abs/2404.19756)
- Liu M, Zeng A, Chen M, Xu Z, Lai Q, Ma L, Xu Q (2022) Scinet: Time series modeling and forecasting with sample convolution and interaction. *Adv Neural Inf Process Syst* 35:5816–5828
- Lobanov V, Firsov N, Myasnikov E, Khabibullin R, Nikonorov A (2024) Hyperkan: Kolmogorov-arnold networks make hyperspectral image classifiers smarter. arXiv preprint [arXiv:2407.05278](https://arxiv.org/abs/2407.05278)
- Nie Y, Nguyen NH, Sinthong P, Kalagnanam J (2022) A time series is worth 64 words: Long-term forecasting with transformers. arXiv preprint [arXiv:2211.14730](https://arxiv.org/abs/2211.14730)
- Olivares KG, Challu C, Marcjasz G, Weron R, Dubrawski A (2023) Neural basis expansion analysis with exogenous variables: Forecasting electricity prices with nbeatsx. *Int J Forecast* 39(2):884–900
- Salinas D, Flunkert V, Gasthaus J, Januschowski T (2020) Deepar: Probabilistic forecasting with autoregressive recurrent networks. *Int J Forecast* 36(3):1181–1191
- Sherstinsky A (2020) Fundamentals of recurrent neural network (rnn) and long short-term memory (lstm) network. *Physica D* 404:132306
- Sidharth S, Keerthana A, Gokul R, Anas K (2024) Chebyshev polynomial-based kolmogorov-arnold networks: An efficient architecture for nonlinear function approximation. arXiv preprint [arXiv:2405.07200](https://arxiv.org/abs/2405.07200)
- Vaca-Rubio CJ, Blanco L, Pereira R, Caus M (2024) Kolmogorov-arnold networks (kans) for time series analysis. arXiv preprint [arXiv:2405.08790](https://arxiv.org/abs/2405.08790)
- Vagropoulos SI, Choularas G, Kardakos EG, Simoglou CK, Bakirtzis AG (2016) Comparison of sarimax, sarima, modified sarima and ann-based models for short-term pv generation forecasting. In: 2016 IEEE International Energy Conference (ENERGYCON), pp. 1–6. IEEE
- Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. *Advances in neural information processing systems* 30
- Wang A, Chen H, Lin Z, Han J, Ding G (2024) Repvit: Revisiting mobile cnn from vit perspective. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 15909–15920
- Wang H, Peng J, Huang F, Wang J, Chen J, Xiao Y (2023) Micn: Multi-scale local and global context modeling for long-term series forecasting. In: The Eleventh International Conference on Learning Representations
- Wang S, Wu H, Shi X, Hu T, Luo H, Ma L, Zhang JY, Zhou J (2024) Timemixer: Decomposable multiscale mixing for time series forecasting. arXiv preprint [arXiv:2405.14616](https://arxiv.org/abs/2405.14616)
- Wang Y, Wu H, Dong J, Qin G, Zhang H, Liu Y, Qiu Y, Wang J, Long M (2024) Timexer: Empowering transformers for time series forecasting with exogenous variables. arXiv preprint [arXiv:2402.19072](https://arxiv.org/abs/2402.19072)
- Weron R (2014) Electricity price forecasting: A review of the state-of-the-art with a look into the future. *Int J Forecast* 30(4):1030–1081
- Wu Y, Guo J, Gopalakrishna G, Poulos Z (2024) Deep-macrofin: Informed equilibrium neural network for continuous time economic models. arXiv preprint [arXiv:2408.10368](https://arxiv.org/abs/2408.10368)
- Wu H, Hu T, Liu Y, Zhou H, Wang J, Long M (2022) Timesnet: Temporal 2d-variation modeling for general time series analysis. arXiv preprint [arXiv:2210.02186](https://arxiv.org/abs/2210.02186)
- Wu H, Xu J, Wang J, Long M (2021) Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Adv Neural Inf Process Syst* 34:22419–22430
- Xu K, Chen L, Wang S (2024) Kolmogorov-arnold networks for time series: Bridging predictive power and interpretability. arXiv preprint [arXiv:2406.02496](https://arxiv.org/abs/2406.02496)
- Yu Y, Si X, Hu C, Zhang J (2019) A review of recurrent neural networks: Lstm cells and network architectures. *Neural Comput* 31(7):1235–1270
- Zeng A, Chen M, Zhang L, Xu Q (2023) Are transformers effective for time series forecasting? In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, pp. 11121–11128
- Zhou C, Huang Y, Kong Y, Lu X (2025) Enhancing trading strategies by combining incremental reinforcement learning and self-supervised prediction. *Expert Syst Appl* 289:128297
- Zhang Y, Yan J (2023) Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In: The Eleventh International Conference on Learning Representations
- Zhang T, Zhang Y, Cao W, Bian J, Yi X, Zheng S, Li J (2022) Less is more: Fast multivariate time series forecasting with light sampling-oriented mlp structures. arXiv 2022. arXiv preprint [arXiv:2207.01186](https://arxiv.org/abs/2207.01186)
- Zhou T, Ma Z, Wen Q, Wang X, Sun L, Jin R (2022) Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In: International Conference on Machine Learning, pp. 27268–27286. PMLR
- Zhou Q, Pei C, Sun F, Han J, Gao Z, Pei D, Zhang H, Xie G, Li J (2024) Kan-ad: Time series anomaly detection with kolmogorov-arnold networks. arXiv preprint [arXiv:2411.00278](https://arxiv.org/abs/2411.00278)

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